

MK HEALTH TRACKER V3

MASTER BUILD DOCUMENT

Owner: Mikail Kocak

1. PROJECT PURPOSE

MK Health Tracker V3 is a personal medical monitoring platform designed exclusively for Mikail Kocak.

The objective is to centralize:

- Diabetes management
- Glucose monitoring
- HbA1c tracking
- Kidney monitoring
- Lipid monitoring
- Medication management
- Medical document storage
- AI medical assistant
- Longevity research news
- Historical reporting

This is a SINGLE USER application.

No registration.

No login.

No multi-user support.

2. PRIMARY HEALTH OBJECTIVES

Priority #1 Kidney Protection

Metrics:

- ACR
- eGFR
- Creatinine

- Blood Pressure

Priority #2 Diabetes Control

Metrics:

- Glucose
- HbA1c
- GMI
- Time In Range

Priority #3 Cardiovascular Risk

Metrics:

- LDL
- HDL
- Triglycerides
- Cholesterol

Priority #4 Longevity

Metrics:

- Weight
- Sleep
- Nutrition
- Supplements

3. MAIN APPLICATION MODULES

1. Dashboard
2. Glucose Tracker
3. Labs & Medical Records
4. Medication Center
5. AI Doctor
6. Reports
7. News Center

4. DASHBOARD REQUIREMENTS

The dashboard is the first page.

Display:

Health Score

Current Glucose

Estimated HbA1c

ACR

LDL

Triglycerides

Blood Pressure

Weight

Display trend charts:

24 Hours

7 Days

30 Days

12 Months

5. GLUCOSE TRACKER

Manual Entry

Photo Upload

PDF Upload

CSV Upload

CGM Upload

Supported Sources:

- Stelo
- Dexcom
- Libre

Calculations:

Average Glucose

Estimated HbA1c

GMI

Time In Range

Time Above Range

Time Below Range

Glucose Variability

6. LABS MODULE

Accepted Files:

PDF

PNG

JPG

DOCX

XLSX

The system must automatically:

Read the file

Extract values

Store results

Generate summary

Generate recommendations

Generate status colors

Green

Yellow

Red

7. REQUIRED LAB VALUES

HbA1c

Glucose

Creatinine

eGFR

ACR

BUN

LDL

HDL

Triglycerides

Total Cholesterol

Vitamin D

TSH

Insulin

Uric Acid

Blood Pressure

Weight

BMI

8. MEDICATION CENTER

Morning

Noon

Night

Medication Name

Dose

Instructions

Reminder Time

Completion Tracking

9. AI DOCTOR

Provider:

OpenAI GPT-5

Capabilities:

Analyze Lab Results

Compare Historical Results

Analyze Glucose Trends

Review Medications

Generate Reports

Answer Health Questions

Generate Doctor Visit Summaries

Important:

Always remind user to consult physician before treatment changes.

10. NEWS CENTER

Update Frequency:

Every 2 Hours

Categories:

Diabetes

Kidney Health

Cardiology

Longevity

Nutrition

Supplements

Sources:

NIH

ADA

NIDDK

Kidney Foundation

Nature Aging

Longevity Technology

11. PDF REPORTS

Generate:

Weekly Report

Monthly Report

Quarterly Report

Annual Report

Include:

Charts

Lab Results

Glucose Trends

Medication List

AI Summary

12. DATABASE TABLES

glucose_entries

lab_documents

lab_values

medications

blood_pressure

weight_logs

health_reports

news_articles

ai_chat_history

13. STORAGE BUCKETS

labs

cgm

reports

images

14. DEPLOYMENT TARGET

GitHub

Vercel

Supabase

OpenAI

15. DEVELOPMENT RULE

Nothing is uploaded to GitHub until:

1. Project builds successfully.
2. npm run build completes without errors.
3. Environment variables are configured.
4. Supabase connection is tested.
5. OpenAI connection is tested.

Only after all checks pass should the first GitHub push be performed.